

[Skip to main content](#)

[Gifts](#)

[Donate](#)



[Our work](#)

[Science](#)



© Frida

Artificial intelligence and conservation

Harnessing the power of technology to protect nature

Skip to main content ce (AI) is transforming how we understand and protect life on Earth. At WWF, we are applying AI tools to help solve some of the most pressing environmental challenges faster, smarter, and at a greater scale.

From identifying wildlife in millions of camera trap images to predicting deforestation before it happens, AI enables us to generate insights from vast and complex data. These insights help scientists, governments, and communities make informed decisions to conserve species and habitats more effectively.

AI doesn't replace traditional conservation—it strengthens it. By combining decades of ecological knowledge with new technology, we can better monitor ecosystems, anticipate threats, and take timely action to protect people and the planet.



Ranger anti-poaching unit testing and training the newly installed mobile FLIR camera system at the Mara Conservancy at Maasai Mara National Reserve Kenya.

© WWF-US / James Morgan

[Skip to main content](#)

ers



AI helps us predict environmental risks, such as wildfires or illegal logging, before they escalate.

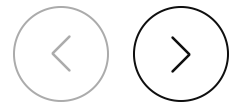
© NOAA



AI helps us support communities with too faster decision-making and resource protection.

© Jasper Doest / WWF

1 / 3



What WWF is doing

[Skip to main content](#)

© Marizilda Cruppe / WWF-UK

By integrating AI into our work, WWF is helping build a more resilient future for people and nature.

WWF is pioneering AI-driven solutions across multiple conservation landscapes. We work with tech companies, local partners, Indigenous peoples, and academic institutions to ensure our tools are grounded in science, equity, and real-world impact. Through innovation and collaboration, WWF is:

Protecting important places

Healthy ecosystems play a critical role in reducing the impacts of climate change. WWF is using AI to protect and restore key ecosystems like mangroves and forests.

Through [ManglarIA](#), “AI for Mangroves” in Spanish, an innovative project supported by Google.org, we’re working to better understand how mangrove ecosystems and their contributions to communities are affected by climate

Skip to main content en use this information to better manage and conserve the long-term viability of mangroves as a nature-based solution for climate change.

Forest Foresight is a WWF initiative that harnesses AI and satellite imagery to detect early warning signs of deforestation and forest degradation. By analyzing patterns in forest health, land use, and human activity, Forest Foresight helps conservationists, governments, and communities act before irreversible damage occurs. This predictive approach enables smarter, faster interventions.

Advancing the responsible use of AI

With decades of experience leading global efforts to protect nature, WWF experts bring a deep understanding of ecosystems, data, and field realities that help guide the responsible application of AI tools. As a trusted partner to governments, Indigenous peoples, local communities, academic institutions, and the private sector, WWF helps ensure that AI innovations are not only technically robust but also grounded in equity, ethics, and real-world impact. This includes considering who has access to and control over data, supporting efforts to uphold Indigenous data rights, and examining the environmental footprint of emerging technologies. WWF is actively working to understand and reduce its own impact, with the goal of using AI in ways that benefit both people and the planet.

Identifying and monitoring wildlife

Monitoring the planet's most at-risk species is critical to their protection. Each year, WWF staff deploy thousands of camera traps—cameras triggered by infrared motion sensors—offering an unprecedented view of wildlife populations and habitat.

Skip to main content and analyzing millions of camera trap photos can take months, delaying the delivery of crucial information to decision-makers. That's why WWF has partnered with leading conservation organizations and Google to develop Wildlife Insights. This innovative platform allows NGOs, governments, and citizen scientists to upload, analyze, and share camera trap data across the conservation community. Powered by advanced AI, Wildlife Insights can automatically identify thousands of species in minutes and allows users to quickly analyze the data to understand where and why wildlife populations are changing.

With easy access to reliable and timely wildlife data on any device, whether in the office or field, Wildlife Insights helps us collaborate more effectively and make more informed and efficient conservation decisions. The Wildlife Insights AI model, called SpeciesNet, is freely available online for anyone to use and improve on.

Stopping wildlife crime

Since 2016, WWF and Teledyne FLIR have collaborated to deploy FLIR thermal cameras to stop rhino poaching in some of Kenya's highest-priority wildlife reserves. The cameras are equipped with night vision and AI that can detect human, wildlife, or vehicle movements within the coverage area in the dark. The system then sends an alert, in real-time, to the operator of the camera system, triggering the appropriate response.

FLIR cameras have greatly increased the effectiveness of nighttime patrols and improved ranger safety. Currently, 11 rhino sites in Kenya are using FLIR cameras and equipment in some capacity, significantly reducing—and in some areas, completely halting—wildlife poaching.

Fighting wildlife trafficking at ports

Skip to main content often exploit air and sea transport to smuggle illegal wildlife and their products, concealing them within large volumes of legitimate baggage and cargo. International airports already use advanced scanning technologies to screen for security threats such as weapons or explosives. Now, WWF and partners are working to deploy similar automated systems to combat wildlife trafficking at key transport hubs.

These wildlife autodetection systems are equipped with sophisticated machine learning algorithms that can automatically identify and flag illegal wildlife products, helping to ensure that nothing is overlooked and that no bad actor can look the other way. By enhancing detection accuracy, reducing the burden on human inspectors, and increasing screening

Convening a community of practice

The AI for Conservation group on WILDLABS, the global conservation technology network, serves as a vibrant hub for conservation practitioners and researchers to share tools, crowdsource answers, find collaborators, discover events, and more. With over 1,000 active members, the group offers a community of support and a network of resources to anyone who uses or is interested in using AI for their conservation work. Whether you're capturing camera trap images, deploying remote sensors, or building machine-learning tools for biodiversity monitoring, this community provides peer support, expert advice, and collaborative opportunities to harness AI responsibly and effectively for conservation impact.

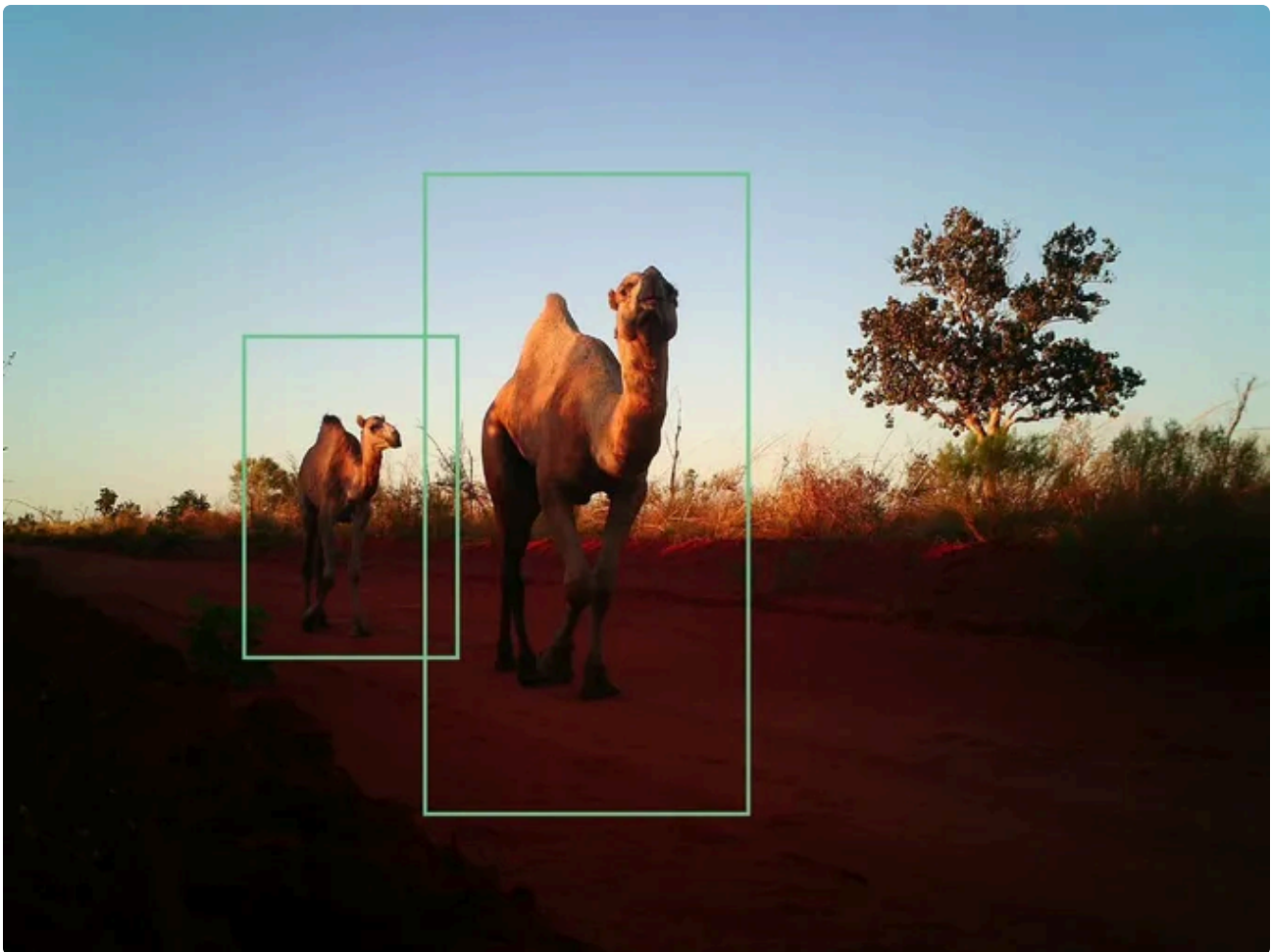
Keeping on top of news

The news and other regularly updated internet-based information sources regularly provide insights into events that may impact WWF's work. AI can help identify these sources of information to enable quicker responses to potential threats and feedback on public sentiment. For example, the

Skip to main content ct provides WWF-India and WWF-Nepal with information about infrastructure construction projects that may impact WWF's areas of interest.

WWF is also developing and piloting SQUID, a low-cost media analysis tool that combines human expertise with AI to monitor sentiment around key conservation issues. By processing insights from news and media sources, SQUID helps track how public perceptions are evolving over time. This can support more responsive and informed strategies across WWF's work.

News and stories

[View all](#)

© Wildlife Insights

Using the power of AI to identify and track species

Skip to main content ie SpeciesNet model saves time and saves wildlife

March 3, 2025

Projects

Wildlife Insights

ManglarIA: Using artificial intelligence to save mangroves in a changing climate

Countering Wildlife Trafficking Through Automated Detection at Ports

Wildlife Crime Technology Project

Experts



[Skip to main content](#)

Vice President, Wildlife Crime and Policy,
Wildlife Conservation



Abby Hehmeyer

Lead, Biodiversity Monitoring &
Conservation Technology, Wildlife
Conservation



Arnaud Lyet

Lead Specialist, Wildlife Conservation

Eric Becker

Conservation Engineer, Wildlife
Conservation Program



Colby Loucks

Vice President and Deputy Lead, Wildlife
Program



Dave Thau

Global Data and Technology Lead
Scientist, Global Science

[Skip to main content](#)

Discover

[Who we are](#)

[Financials](#)

[Latest stories](#)

[Our work](#)

[En español](#)

[Newsroom](#)

[Travel](#)

Connect

[Take action](#)

[Careers](#)

[Contact](#)

[Help center & FAQs](#)

[WWF country offices](#)

[Ethics reporting](#)

[Safeguards redress mechanism](#)

Support

[Donate now](#)

[Visit WWFGifts](#)

[Give in someone's name](#)

[Gift and estate planning](#)

[Skip to main content](#)

Get WWF email and texts

Supporter log in 



World Wildlife Fund

1250 24th St NW

Washington, DC 20037

World Wildlife Fund Inc. is a nonprofit, tax-exempt charitable organization (tax ID number 52-1693387) under Section 501(c)(3) of the Internal Revenue Code. Donations are tax-deductible as allowed by law.

© 2026 World Wildlife Fund. WWF® and ©1986 Panda Symbol are owned by WWF. All rights reserved.

Site Terms**Privacy Policy****Your Privacy Choices****State Disclosures****Accessibility Statement****Cookie settings**